The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies

Erik Brynjolfsson and Andrew McAfee (both MIT Center for Digital Business).
NY: W. W. Norton, Jan 2014, 306p, $26.95. (www.secondmachineage.com)

On the surface, this is a very important book about present and future technologies, jobs, and growing inequality. It is clearly written, plausible, and well-documented. Although oriented to American audiences, it has global import in our globalizing age. It is not directly about security and sustainability (neither term is in the index), although the book could illuminate both concerns, as technology continues its inexorable advance, for better *and* worse.

The first machine age brought the Industrial Revolution: the sum of several near simultaneous developments in mechanical engineering, chemistry, metallurgy, etc. The most important technology was the steam engine, which overcame limitations of human and animal muscle power, leading to factories, mass production, railways, and mass transport.

"Now comes the second machine age. Computers and other digital advances are doing for mental power—the ability to use our brains to understand and shape our environment—what the steam engine and its descendants did for muscle power" (pp.7-8). How this transition will play out is unknown, but it is "a very big deal," for mental power is at least as important for progress and development as physical power.

1. Three Broad Observations

1. ***Rapid Progress Ahead.*** "We're living in a time of astonishing progress with digital technologies"— those that have computer hardware, software, and networks at their core. Just as it took generations to improve the steam engine, it has also taken time to refine our digital engines. The full force of these technologies has only recently been achieved, yet computers will continue to improve and do new and unprecedented things. "Full force" simply means that key building blocks are now in place. In short, "we're at an inflection point" where the curve starts to bend a lot.
2. ***Profound Benefit.*** "The transformations brought about by digital technology will be profoundly beneficial ones." The new era will be better because we'll be able to increase the variety and the volume of our consumption. "Technology can bring us more choice and even freedom," and technical progress is improving exponentially.
3. ***Tough Challenges.*** Digitization will bring with it some thorny challenges, notably economic disruption, because, as computers get more powerful, employers will have less need for some kinds of workers. "*Technological progress is going to leave behind some people, perhaps even a lot of people, as it races ahead."* (p.11, italics added). The challenges of the digital revolution can be met, but we first have to be clear on what they are, and to discuss the likely negative consequences. They're not insurmountable, but they won't fix themselves.

2. Recent Technological Progress

Six chapters enthusiastically discuss the skills of the new machines (self-driving cars from Google's Chauffeur project, instantaneous translation from IBM's GeoFluent and Google Translate, acceleration in robotics, 3D printing used by many companies to make prototypes and model parts), why Moore's Law has held up for so long (brilliant tinkering leading to constant modification), the digitization of just about everything, the fundamental importance of innovation for growth and prosperity, why innovation and productivity will continue to grow at healthy rates in the future, the life-changing potential of Artificial Intelligence (bringing key aspects of sight to the visually impaired, restoring hearing to the deaf, AI-aided diagnoses in some medical specialties).

Echoing the optimism of the late Julian Simon, the authors exclaim that "there is no better resource for improving the world and bettering the state of humanity than the world's humans… Our good ideas and innovations will address the challenges that arise, improve the quality of our lives, allow us to live more lightly on the planet, and help us take better care of one another. It is a remarkable and unmistakable fact that, with the exception of climate change, virtually all environmental, social, and individual indicators of health have improved over time, even as human population has increased" (p.93). But the main impediment to more progress has been that, until quite recently, many people had no effective way to access the world's knowledge. This situation is rapidly changing with the advent of mobile phones, bringing billions of people into the community of potential knowledge creators, innovators, and problem-solvers.

3. Bounty and Spread

The next five chapters explore the two economic consequences of this progress: *bounty* (the increase in volume, variety, and quality and the decrease in cost of the many offerings of modern technology) and *spread* (the ever-bigger differences among people in income and wealth, likely to accelerate unless we intervene). Topics include marked increases in productivity growth, zero-price products and services not reflected in GDP (e.g. over one million apps on smartphones, Wikipedia, free classifieds on Craigslist, free phone calls on Skype), the ubiquitous bounty of digital photographs, intangibles as a growing share of capital assets (intellectual property, organizational capital, user-generated content, and especially human capital), and the need for "new metrics" other than GDP in the second machine age.

But growing inequality or spread is a major problem. The authors discuss decoupling of median wages from productivity (the bottom 80% of the US income distribution saw a net decrease in their wealth since 1983), increased earnings of the top 1% by 278% between 1979 and 2007 while overall median income has fallen since 1999, economic winners and losers ("digital technologies increase the economic payoff to winners while others become less essential and hence less well rewarded"), the evolving skill set affected by computerization, stars and superstars as the biggest winners due to "winner-take-all" markets (the top 0.01% saw their share of national income double from 3% to 6% between 1995 and 2007), why winner-take-all markets are more common now (digital goods have enormous economies of scale), the questionable "strong bounty" argument that it will overwhelm the spread and thus no need to worry ("we wish that were the case, but it's not"), technological unemployment despite a growing economy (but most mainstream economists still argue that technology creates more jobs than it destroys), and globalization. *"In the long run, the biggest effect of automation is likely to be on workers not in America and other developed nations, but rather in developing nations that currently rely on low-cost labor for their competitive advantage"* (p.184). The advantage of low wages largely disappears by installing robots and other types of automation; "offshoring is often only a way station on the road to automation."

4. Labor Force Remedies

1. ***Individuals.*** "Our most fundamental recommendations to students and their parents: study hard, using technology and all other available resources to 'fill up your toolkit' and acquire skills and abilities that will be needed in the second machine age" (p.199). Learn to race with machines. A college degree remains a vital stepping stone to most careers. Most professions still require the skills of ideation, large-frame pattern recognition, and complex communication. "As the labor market polarizes more and the middle class continues to hollow out, people who were previously doing mid-skill knowledge work start going after jobs lower on the skill and wage ladder…this puts downward pressure on wages" (p.202). More surprises are in store, and "it's becoming harder and harder to have confidence that any given task will be indefinitely resistant to automation" (p.203).
2. ***National Policies.*** The best way to tackle labor force challenges is to grow the economy. To do so, we should teach children well, put digital technology to work (e.g. MOOCs enable low-cost replication of the best teachers, content, and methods), "flip the classroom" by having students listen to lectures at home and do traditional homework at school, raise teacher salaries, lengthen school hours and the school year, champion the innovation engine of entrepreneurship (the best way to create jobs and opportunity), improve matches between jobs and people, support basic research, institute prizes for innovation, upgrade US infrastructure to acceptable levels (one of the best investments the country can make), reform counterproductive immigration policy (of benefit not only to immigrants but to the economy), tax wisely (most economists advocate "Pigovian" taxes on pollution and other negative externalities; also, a value-added tax and higher taxes on high earners).
3. ***Long-Term Policies.*** Reward employment instead of taxing it; some form of basic income guarantee can help, but better still is a negative income tax that provides an incentive to work. Also, we will need some radical "out-of-the-box" ideas, e.g.:

– A national mutual fund distributing ownership of capital widely;

– Directing technical change toward machines that augment human ability rather than substitute for it;

– Paying people to do socially beneficial tasks;

- Nurturing special categories of work to be done by humans only (e.g. caring for the young and old);
- Awarding credits to companies that employ humans, similar to carbon offsets that can be purchased;
- Providing vouchers for basic necessities like food, clothing, and housing;
- Ramping up government programs like the depression-era Civilian Conservation Corps to clean up the environment and build infrastructure.

[NOTE: Not mentioned but well worth adding to this list are job-sharing programs that match the over-worked with the under-worked, promoting quality part-time work with better benefits (another form of job-sharing), and perhaps encouraging semi-self-sufficiency, e.g. with urban agriculture programs and homesteading. See **Work Sharing During the Great Recession;** International Labor Office, May 2013.]

"Technology is not destiny. We shape our destiny."
– Erik Brynjolfsson & Andrew McAfee

---

5. Other Long-term Prospects and Problems

The final chapter emphasizes that the new technologies are exponential and combinatorial, with most of the gains still ahead. Growing even faster than Moore's Law, "in the next 24 months, the planet will add more computer power than it did in all previous history; over the next 24 years, the increase will likely be over a thousand-fold" (p.251). Our generation will likely experience "two of the most amazing events in history: the creation of true machine intelligence and the connection of all humans via a common digital network, transforming the planet's economics." But not all the news is good, because, *"while the bounty brought by technology is increasing, so is the spread."* And this is not the only possible negative consequence.

As we move deeper into the second machine age, "perils from both accident and malice will become greater, while material wants and needs are likely to be relatively less important." We will be increasingly concerned with questions about catastrophic events, genuine existential risks, freedom vs. tyranny, etc. The sheer density and complexity of our digital world bring risks and weaknesses: 1) the digital world is subject to minor initial flaws that cascade via an unpredictable sequence into something much larger and more damaging; 2) tightly-coupled complex systems make tempting targets for spies, criminals, and terrorists ("until recently, our species did not have the ability to destroy itself; today it does"); 3) there are myriad other ways that technology can have unexpected side effects, e.g. addictive gambling, digital distractions, cyber-balkanization of interest groups, social isolation, and environmental degradation; 4) development of fully conscious machines may bring a dystopian "terminator" future, or a utopian singularity; "we honestly don't know; as with all things digital, it's wise never to say never, but we still have a long way to go." (p.255)

"In the second machine age, we need to think much more deeply about what it is we really want and what we value, both as individuals and as a society… Technology is not destiny. We shape our destiny" (Final paragraph, p.257).

COMMENT: IS THIS "REALLY BIG DEAL" WHAT WE NEED?

The Second Machine Age has much in common with several other recent "techno-ecstatic" books on the new digital technology, notably **The New Digital Age** by Eric Schmidt and Jared Cohen, **Big Data: A Revolution That Will Transform How We Live, Work, and Think** by Viktor Mayer-Schonberger and Kenneth Cukier, and **Abundance: The Future Is Better Than You Think** by Peter Diamandis and Steven Kotler (GFB Book of the Month, Aug 2012). All four books promise a better world ahead. Diamandis and Kotler make no qualifications at all, but simply enthuse that, within a generation "abundance for all is actually within our grasp," with no mention of how it will be distributed. The promise of "big data" is hyped in the book's sub-title, with no downsides of infoglut considered. "Googlers" Schmidt and Cohen state that everyone will benefit in the new digital age, but not equally— which is a slight improvement in sobriety.

The importance of Brynjolfsson and McAfee is that they forthrightly venture into the economics of the new digital era, linking the new technologies to the widely-appreciated fact of growing inequality. Although they state that the transformations ahead will be "profoundly beneficial," they also grapple with the "thorny challenges" of technological unemployment and uneven "spread" of the "bounty," with two chapters of suggestions on how to tackle labor force challenges. The authors state that we *can* face up to these challenges, even as the spread accelerates—which they deem likely. But, seriously, folks, how likely is it that Americans and other developed nations will do so to any meaningful degree? Similarly, the Worldwatch Institute says that sustainability is still possible if we do everything right starting now (**Is Sustainability Still Possible?** GFB Book of the Month, Oct 2013), but how likely is that?

The authors mention only in passing that the biggest effect of technological displacement of jobs will be on the developing nations (p.184), which parallels the unfortunate megatrend that the biggest negative impacts of climate change, largely caused by the rich nations, will also impact the poorest nations. And they don't consider the declining quality of many jobs as a result of economic restructuring (e.g., see David Weil, **The Fissured Workplace: Why Work Becomes So Bad For So Many**), or the impact of the new technologies on politics. **The New Digital Age** does consider politics, noting more "revolutions" ahead for better or worse by newly enlightened and connected citizens—especially angry young people who are unemployed or underemployed as a result of the new technologies. We can already see roiling civic discontent in a dozen or so countries worldwide, which does not necessarily lead to better leadership, policies, or jobs (e.g., see **Global Employment Trends for Youth: A Generation at Risk;** International Labor Office, Aug 2013).

Brynjolfsson and McAfee conclude that "we need to think much more deeply about what it is we really want and really value." We *also* need to think more deeply, widely, and rigorously about current social, economic, and environmental trends and possible developments, as well as the "brilliant technologies" that are driving them. How much and what kind of bounty are we really getting, and at what cost to whom, all things considered? We need nutritious food, potable water, safe and affordable energy, decent housing, adequate infrastructure, and security in many dimensions. But will the Second Machine Age provide more than marginal help in these essential areas? These basic needs are not addressed by Brynjolfsson and McAfee, who focus instead on the wonders of self-driving cars (perhaps important to those traveling to Silicon Valley on California's congested highway 101) and on the flood of information now available worldwide (but not necessarily utilized, or beneficial, and too often a distraction).

The authors make a strong case that "a very big deal" is unfolding—that "we're at an inflection point" where the curve starts to bend a lot. This may be overstated to some degree, but should not be ignored. Similarly, environmental scientists have been warning of various "tipping points" and "abrupt climate changes" ahead (e.g., **Abrupt Impacts of Climate Change;** GFB Book of the Month, Jan 2014). The difference in presentation is that the scientists are restrained in their assessments, often erring on the side of caution and trying not to express undue pessimism. In contrast, those who write about technological change are too often exuberant and overly optimistic. To its credit, **The Second Machine Age** does devote an entire chapter ("Beyond GDP," pp 107-124) to discussing the deficiencies of the obsolete GDP measure of production and progress. It just doesn't go far enough. "True growth is greater than the standard data suggest." (p.119) But if the negatives are subtracted, overall net growth may be much less.

Author Contact Information
Email: mmarien@twcny.rr.com

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.